

MAG — every variable, what it means, and why its weight is what it is

2026-07-27 · 47 features · fitted on 82,836 human decisions from 3,340 clean open-sheet games

GENERATED by `build/build_mag_variables.js` from the live weight file. Do not hand-edit the tables.

Nobody typed these numbers. Each is fitted: the model is shown a real human decision and all the options that person had, and the weights are nudged until the option they actually clicked comes out most likely. So every weight answers one question — **what makes a human more likely to click this** — and never *what wins*.

`ZERO` marks a weight whose 95% interval contains zero: the data cannot tell it from no effect.

The last column is the same fit with **popularity removed entirely**.

On held-out human decisions the full model picks the option the person actually clicked **30.9%** of the time; without popularity, **28.7%** — **a loss** of 2.2 points. So popularity carries real signal and is not crowding the board facts out. An earlier version of this document asserted the opposite, from a fit whose `DROP=` reached only half the corpus.

What the move does

feature	weight	no-popularity	what it means
<code>eff4</code>	+1.277	+1.204	it hits a 4x weakness
<code>eff2</code>	+0.928	+0.894	it hits a 2x weakness
<code>stab</code>	+0.012 <code>ZERO</code>	+0.164	it matches my own type
<code>bp</code>	+0.006 <code>ZERO</code>	+0.003	it is a powerful move
<code>allyHit</code>	-0.043 <code>ZERO</code>	-0.133	it also hits my own partner, and my partner is not immune to it
<code>accuracy</code>	-0.619	-0.662	how often it hits, on THIS board (snow makes Blizzard certain)
<code>effHalf</code>	-0.852	-0.835	it is resisted
<code>effQuarter</code>	-1.004	-1.007	it is resisted twice over
<code>immune</code>	-2.198	-2.155	it does nothing at all
<code>abilityBlock</code>	-2.300	-2.286	the target probably has an ability that eats it

MOVES THAT COST MORE THAN A TURN

feature	weight	no-popularity	what it means
<code>rechargeTurn</code>	-0.443	-0.575	it costs me the turn AFTER this one
<code>chargeTurn</code>	-1.306	-1.266	it needs a turn to wind up on this board

PIVOTS

feature	weight	no-popularity	what it means
isStatus	+0.528	+0.517	it is a status move
tgtHurt	+0.366	+0.346	the target is already hurt
pivots	-0.588	-0.526	it damages and brings me out
deadStall	-1.147	-1.124	I protected last turn, so it would probably fail
deadWeather	-1.185	-1.305	that weather is already set, so it would fail
deadStatus	-1.414	-1.279	the target already has a status, so it would fail
deadField	-1.894	-1.507	that field effect is already up, so it would fail
pranksterFailsDark	-1.920	-1.862	my Prankster status move is aimed at a Dark type, so it does nothing
deadSide	-2.161	-2.016	that side effect is already up, so it would fail
deadNoLastMove	-3.434	-3.392	it needs the target to have already moved, and the target just switched in

WHO MOVES FIRST

feature	weight	no-popularity	what it means
defMismatch	+0.371	+0.362	I hit its softer defence
tgtBulk	+0.186	+0.211	the target is bulky
movesFirst	+0.112	+0.113	I move before the target, counting priority, Tailwind and Trick Room
tgtPhysical	-0.269	-0.265	the target attacks physically rather than specially (negative = specially)
priority	-0.397	-0.287	this move cuts the queue (Fake Out, Sucker Punch, Extreme Speed)

DOES IT KILL

feature	weight	no-popularity	what it means
killIsRoll	+0.941	+1.007	it kills some spreads and not others: a roll rather than a read
tgtMayProtect	+0.574	+0.613	how often this target blocks: the biggest reason a sure kill is not one
protectThreatened	+0.261	+0.285	this move protects, and I am facing a kill
koFirst	+0.060	+0.063	it kills, and I move first, so the kill lands before their attack
killsThreat	-0.064	-0.062	it kills the thing that was about to kill me
dmgFrac	-0.076	-0.053	how much of what is left of the target it takes
koTarget	-0.164	-0.141	the odds this really kills it: the worst roll still does, and the move lands

WHAT A MOVE DOES TO THE OTHER PLAYER'S OPTIONS

feature	weight	no-popularity	what it means
volatileOnSelf	+0.185	+0.234	it puts something on me or my side (Substitute, Follow Me, Rage Powder)
volatileOnFoe	-0.207	-0.280	it takes an option away from the target (Taunt, Encore, Disable)

STAT STAGES

feature	weight	no-popularity	what it means
myOffenseStage	+1.969	+1.995	how boosted the stat I am attacking with already is
movesBoostMe	+0.201	+0.224	this move raises one of my own stats
tgtDefenseStage	-0.011 ZERO	-0.032	how boosted the defence I am attacking into already is
movesLowerFoe	-0.406	-0.885	this move lowers one of the target's stats

STATUS THAT ACTUALLY BITES

feature	weight	no-popularity	what it means
statusBites	+0.531	+0.343	the status this inflicts hits a stat the target actually relies on

AM I EVEN GOING TO GET TO DO THIS

feature	weight	no-popularity	what it means
diesBeforeMoving	-0.432	-0.433	I am facing a kill and I do not move first

SWITCHING

feature	weight	no-popularity	what it means
switchSurvives1	+0.240	+0.232	the replacement lives through the hardest thing aimed at me
switchSurvives2	-0.124	-0.124	it lives through that twice
switchFaster	-0.128	-0.119	the replacement outruns the thing that was threatening me
isSwitch	-0.167 ZERO	-0.885	this candidate is a switch, not a move

WHAT THE CROWD DOES

feature	weight	no-popularity	what it means
priorLogP	+0.157	+0.000	it is a popular move